

Pearson Edexcel International GCSE ICT (4IT1)

Paper 1: Software Systems Mock Examination

Teacher Reference: Tr Jamal Fahadh | **Target Cohort:** ICT Candidates, Best Academy, Hargeisa, Somaliland

Time Allowed: 2 Hours and 30 Minutes | **Total Marks:** 150 Marks

SECTION A: Operating Systems & Core Infrastructure (25 Marks)

Question 1

A logistical distribution company uses a network of computing systems to coordinate its shipping services.

(a) State what is meant by system software. (1 Mark)

Define system software and its role in hardware management...

(b) Identify one example of application software from the list below. Place a cross [X] in the correct box. (1 Mark)

Box	Software Option
☐	A. Operating System
☐	B. Defragmentation Utility
☐	C. Spreadsheet Software
☐	D. Disk Formatting Tool

(c) Contrast the function of a single-purpose computer with a general-purpose computer. (2 Marks)

Contrast hardware adaptability and functional limitations...

(d) The company's administrative office uses a Network Operating System (OS). Describe two functions specific to a Network OS that are not found on a Single-User OS. (4 Marks)

Function 1 & Function 2 details regarding multi-user environments...

(e) Office workers send large invoice documents to a shared office printer simultaneously. Explain how the operating system handles this scenario using print spooling. (3 Marks)

Explain print queues, buffer memory structures, and job serialization...

(f) When an application is launched, the operating system manages memory allocations. Describe the process of memory management when an application is opened and subsequently closed. (3 Marks)

Describe RAM allocation hooks and de-allocation processes...

(g) The company handles thousands of employee profiles and access permissions. Identify three methods an operating system uses to control machine access via its security/authentication function. (3 Marks)

1. Identify first credential validation path... 2. Identify second token or biometric path... 3. Identify third verification layer...

(h) Software procurement parameters govern operational legality across the organization.

(i) Define what is meant by a software licence. (1 Mark)

Define legal usage parameters and permissions terms...

(ii) State the name of the legal violation committed if unlicensed software is deployed across the company network. (1 Mark)

State the explicit legal copyright violation framework...

(iii) Explain three benefits to users when choosing free or open-source software over proprietary applications. (6 Marks)

Benefit 1, Benefit 2, and Benefit 3 text (financial, code modifications, distribution flexibility)...

SECTION B: System Maintenance & Utility Utilities (30 Marks)

Question 2

System maintenance is critical to keeping the distribution centre's hard drives performing optimally.

(a) Define the term utility software. (1 Mark)

Define system-level optimization or house-keeping software utilities...

(b) The company's hard disk drives (HDDs) slow down over extended periods of usage.

(i) Explain why data files on a mechanical hard disk drive become fragmented over time. (2 Marks)

Explain non-contiguous storage allocations and file modifications...

(ii) Describe how a defragmentation utility remedies this issue to improve file loading access times. (3 Marks)

Describe file sector sorting and platter read/write head movement reduction...

(c) Network administrators schedule automated system backups. Explain why backup utilities are typically configured to execute at off-peak times. (2 Marks)

Explain network bandwidth conservation and server load constraints...

(d) Large data reports must be emailed to international offices daily.

(i) State the primary objective of a compression utility. (1 Mark)

State the reduction metric of physical bit size...

(ii) Explain how compression algorithms like Run-Length Encoding (RLE) reduce physical file size. (3 Marks)

Explain repeating data tokens and structural frequency mapping rules...

(iii) State what action must occur before a compressed file can be read directly by its original program. (1 Mark)

State the prerequisite file conversion step...

(e) A set of older external hard drives needs to be re-allocated for new operations.

(i) Describe the function of a disk formatting utility when preparing a drive for its first use. (2 Marks)

Describe track structures, sector initializations, and file systems installation...

(ii) State the warning associated with executing a formatting operation on a drive that already contains data. (1 Mark)

State the explicit absolute data loss warning details...

(iii) Explain why data from a freshly formatted drive can occasionally be recovered by specialists. (2 Marks)

Explain file index clearing parameters vs real data bits persistence...

(f) Explain four distinct reasons why developers push software updates or patches. (8 Marks)

Reason 1, 2, 3, and 4 (security holes, bug fixes, feature extensions, compatibility updates)...

(g) State the preventative system safety rule an operator must follow before launching any critical software update. (2 Marks)

State the standard administrative archival rollback strategy...

SECTION C: Generic Applications Portfolio (35 Marks)

Question 3

A retail business manages inventory datasets, marketing designs, and staff schedules using a wide assortment of generic applications.

(a) State the Exam Tip rule regarding how you must refer to applications when answering Pearson Edexcel exam questions. (1 Mark)

State the rule on avoiding specific software brand names...

(b) Identify one generic application category used specifically to model corporate finances and execute 'what-if' scenario checks. Place a cross [X] in the correct box. (1 Mark)

Box	Generic Software Option
☐	A. Word Processing
☐	B. Spreadsheet Software
☐	C. Database Management Systems (DBMS)
☐	D. Desktop Publishing (DTP)

(c) Complete the table below by matching the generic software type with its primary focus and features. (8 Marks)

Generic Software Type	Primary Focus & Core Features
Word Processing	
Database (DBMS)	
Desktop Publishing (DTP)	
Web Authoring	

(d) The marketing department modifies visual imagery for corporate advertising campaigns. Differentiate between the operational traits of a Graphics Editor and a Photo Editor. Your answer must reference vector graphics, bitmap image structures, shapes, and pixel coordinates. (6 Marks)

Contrast coordinate spaces, scaling fidelity, layer formats, and canvas traits...

(e) The media team relies on Sound Editors to manipulate audio records for promotional broadcasts. State three features provided by sound editing software utilities. (3 Marks)

1. Trimming or clipping details... 2. Volume transformations... 3. Filter deployment features...

(f) Staff members are preparing presentations to back up public speaking slots at a commercial conference. Describe how Presentation Software properties support a presenter during a live event. (2 Marks)

Describe speaker notes layouts, slide transition triggers, or display modes...

(g) Communication software provides multiple remote network connection entry points to transfer data files. State four types of communication services covered within this software classification framework. (4 Marks)

List four standard network communication services covered in the syllabus...

(h) The business schedules building extensions using Project Management Software. Explain two benefits to a business manager of using project management applications to oversee multi-stage operational tracks. (4 Marks)

Benefit 1 & Benefit 2 mapping task dependencies, critical paths, and resource charts...

(i) Automated computer units control infrastructure components.

(i) Identify the generic software type responsible for reading electronic sensor inputs and commanding physical mechanical actuators. (1 Mark)

Identify the control system type classification...

(ii) List two real-world deployments of this software form within automated industries. (2 Marks)

1. Identify first production layout application... 2. Identify second automation use case...

SECTION D: Comparative Performance Matrix (35 Marks)

Question 4

An IT consultant is preparing a structural analysis report evaluating software application profiles for a business launch.

Complete the missing properties within the software classification matrix below by directly sourcing technical criteria from the textbook guidelines. (35 marks, 5 marks per row)

Generic Software Form	Primary Intended Target Purpose / Features	Sourced Application Example
Word Processing	Text-based documents; features grammar checkers and layout formatting.	(a)
Desktop Publishing	(b)	Adobe InDesign
Spreadsheets	(c)	Microsoft Excel
Databases (DBMS)	Structured data storage; features data entry forms and query editors.	(d)
Web Authoring	(e)	Adobe Dreamweaver
Graphics Editors	(f)	Serif Affinity Designer
Photo Editors	Modifies bitmap image grids; features cropping and red-eye removal.	(g)